

Clean Document Extraction — Accuracy Report

Capstone Project — accuracy evaluation of the LLM PDF parser

1. What was tested

Forty clean documents across six types, each with hand-prepared ground truth, parsed by a local Ollama model with a deterministic extraction and enrichment layer around it.

Document type	Count	Ground truth available
Resumes	10	Field-level JSON
Receipts	10	Field-level JSON
Invoices	10	Field-level JSON
Application forms	9	Field-level JSON
Contracts	5	Verbatim page transcription
Reports	5	Verbatim page transcription

The two ground-truth formats matter, and are the reason three different metrics appear in this report (Section 5).

2. Headline results

Document type	Count	Metric	Result
Receipts	10	Field accuracy	100% (70/70)
Invoices	10	Field accuracy	100% (50/50)
Contracts	5	Content recall	99.4%
Resumes	10	Per-field scorer	93.9%
Forms — typed	6	Value recall	88.1%
Reports	5	Content recall	85.6%
Forms — handwritten	3	Value recall	13.8%
Document classification	40	Correct type	100% (40/40)

Every document was routed to the correct parser. Receipts and invoices reach 100% because those fields are extracted deterministically and no longer depend on the language model at all.

3. Improvement over the project

Area	Starting point	Final
Receipts	Crashed on upload (HTTP 500)	100%
Invoices	82.9%	100%

Area	Starting point	Final
Contracts	17.9% content recall	99.4%
Reports	3.5% content recall	85.6%
Resumes	90.8%	93.9%
Classification	3 document types misrouted	40/40

4. Results by document

4.1 Receipts — 100% (70/70)

Merchant name, receipt number, subtotal, tax, total, currency and payment method verified against `ground_truth_receipts.json` for all ten documents.

These fields are read directly off the document by label matching and payment-line parsing. The result was measured starting from an **empty model response**, which demonstrates that the model contributes nothing to these particular fields — the figure is reproducible rather than sampled.

4.2 Invoices — 100% (50/50)

Invoice number, subtotal, tax, total and currency verified against `ground_truth_invoices.json` for all ten documents.

4.3 Contracts — 99.4%

Contract	Content recall
Digital Marketing Proposal	100.0%
Financial NDA	100.0%
Buyer Agency Agreement	99.2%
Car Rental Agreement	99.2%
Internship NDA	98.3%

4.4 Resumes — 93.9% average

Resume	Score
sebastian_bennett	98.2%
hannah_morales	98.1%
lorna_alvarado	97.1%
daniel_gallego	96.4%
nelly_hassan	94.8%
jonathan_patterson	93.5%
donna_stroupe	93.3%
juliana_silva	90.0%

Resume	Score
henrietta_mitchell	89.1%
sharya_singh	88.9%

Part of henrietta_mitchell's shortfall is not recoverable: her ground truth pairs job dates with the wrong employers. This was confirmed by rendering the page to an image and reading it — the parser matches the document, and the ground truth does not.

4.5 Reports — 85.6%

Report	Content recall
OUWRPS_15_05_published	99.3%
Different_Mining_Techniques	99.2%
Sep2021-SCIFuzzy	96.7%
automotive-8d-demo	92.4%
Low_intensity_behavioral	62.2%

Low_intensity_behavioral is limited by PDF text extraction rather than by parsing. Only **63.1%** of that document's vocabulary is recoverable from its text layer at all, and the parser captured 61.6% — **98% of what is achievable**. Excluding it, reports average approximately 97%.

4.6 Application forms

Form	Recall	Type
passport_application_meridale	94.7%	typed
loan_application_coral_bay	89.7%	typed
college_application_ashford_vale	89.5%	typed
loan_application_summit_horizon	88.9%	typed
college_application2	85.0%	typed
college_application1	78.0%	typed
passport	18.5%	handwritten
job_application	16.9%	handwritten
job	3.3%	handwritten

The typed/handwritten split is the meaningful result. Handwritten forms are capped by OCR, not by parsing: only **25–41%** of their ground-truth values survive into the extracted text. The vision model reads some fields correctly and others confidently wrong — one date was read as 12/01/99 instead of 12/01/79, which is more dangerous than a blank because nothing downstream flags it.

5. Why three different metrics

The metrics are not interchangeable and should not be averaged into a single figure.

Metric	Applied to	Question it answers
Field accuracy	Receipts, invoices	Does each named field match the expected value exactly?
Value recall	Forms, resumes	What fraction of ground-truth values appear in the output?
Content recall	Contracts, reports	What fraction of the document's substance reaches the output?

Ground truth for contracts and reports is a verbatim page transcription rather than a set of expected field values, so field-level accuracy cannot be computed for those types.

Precision and recall diverge sharply

For forms the two measures give very different answers on the same output:

Measure	Result
Precision — of the fields produced, how many are correct	94.1%
Recall — of everything expected, how much was captured	69.7%
F1	80.1%

Leaf-name matching alone reports ~94% because a field the parser **omitted** has no counterpart to compare against and silently drops out of the denominator. 230 of 469 ground-truth values were excluded that way, and 128 of those were genuinely absent from the output. Quoting precision alone implies a completeness the parser does not have.

6. How the accuracy was achieved

6.1 Moving structured fields off the model

The largest gains came from a change in approach rather than in prompting. For receipts and invoices, the important values are printed beside unambiguous labels. A regex that matches "Subtotal" and takes the adjacent amount is strictly better than a model call, because a regex match either is or is not literal source text — there is no hallucination risk in what it captures.

Moving subtotal, tax, total, currency and payment method to deterministic extraction resolved several real errors at once:

- A grocery receipt reported `payment_method`: "Cash" when the payment line read "Visa Contactless ** 1842" — the model had been misled by "Cashier 06" earlier in the document.
- Tax was returned as "13%" — the *rate* rather than the *amount* — because the tax label list omitted **HST**, so Canadian tax lines fell through to the model.
- A restaurant bill returned the pre-tip total, because the label helper returned whichever label appeared first in the document rather than respecting the intended priority order.

6.2 Reading order

The original logic classified a whole page as either one-column or two-column. Real pages are not uniform: one resume runs full width at the top and splits into two columns lower down, and a single global decision gets it wrong either way. A recursive **XY-cut** now decides locally — looking for a clean vertical gap, then a clean horizontal one, and recursing.

Two related fixes came from the same work. Long URLs that wrap mid-token in a narrow sidebar are now rejoined deterministically, instead of being reassembled by the model with a separator that was never in the document. And a vertical split is refused when the two bands have matching row counts at matching y-positions, because that pattern indicates a label/value table rather than genuine columns — the original XY-cut had made receipt totals worse by separating labels from their amounts.

6.3 Contracts and reports

Both initially discarded almost all of their content, returning section headings with no text beneath them. A prompt change requiring {heading, content} pairs did not work: the type-specific guidance sits roughly 3,000 characters into the prompt with up to 24,000 characters of document text after it, and instructions that far from the generation point are ignored.

The working solution was to stop relying on the prompt. A contract's clauses and a report's sections are **layout** — numbering and heading lines — and can be segmented deterministically. The segmenter handles numbered clauses (1., 3.2, ARTICLE IV), heading-labelled sections, and unnumbered prose, since many NDAs have no numbering at all.

Contracts moved from 17.9% to 99.4%; reports from 3.5% to 85.6%.

7. Where the measurement was wrong

Three times an apparent accuracy problem turned out to be a measurement problem. These materially affect how the numbers should be read.

Addresses scored 0% while being correct. Three resumes scored zero on `contact.address`:

```
ground truth '43-589 Beechwood Dr, Waterloo, ON N2T 2K9'
extracted    '43-589 BEECHWOOD DR WATERLOO ON N2T 2K9'
```

Identical content; only case, commas and spacing differ. Exact string equality was grading transcription style rather than whether the field was found. This moved the resume average from 90.8% to 93.9% **without changing any extraction code**.

Reports appeared to score 98%. That figure measured whether extracted values appear in the source — but the parser was emitting only section headings, and headings do appear in the source. Measured as content recall, the true figure was 3.5%. Precision looked excellent while the substance of every report was being discarded.

A rule was written backwards from the test data. While fixing invoice currency detection, a rule inferring CAD from the presence of HST was added — reasoning from a Canadian test set rather than from the document. Investigating properly exposed a worse latent bug: the amount pattern recognised only \$, € and £ and assumed digits group in threes, so an Indian invoice's Rs.1,41,760.00 extracted as 41,760.00 — a figure 3.4x too small, silently.

8. Limitations

Results come from several runs. Receipts, invoices, contracts and reports were measured on one build; forms on a later one; resumes earlier still. A single clean re-run would make the set internally consistent.

Model output is not deterministic. Two runs of identical code on the same forms produced per-document swings of up to ±13 points while the aggregate stayed at 69.7%. Single-run figures should be treated as samples rather than measurements; a median of three runs would be more defensible.

Resume scoring uses a corrected metric. The same extraction output scored 90.8% under the original scorer and 93.9% under the corrected one. Extraction quality is identical; only the measurement changed.

Handwritten forms are hardware-limited. The gap is OCR and vision-model capability on a memory-constrained machine, not parser logic.

One ground-truth file contains errors. henrietta_mitchell's experience dates are paired with the wrong employers, and the education/activities organisations are swapped. Verified by rendering the page. Her score is therefore understated.